National Geo-AI Hackathon: Theme 1 Report

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1. Executive Summary

This project aims to automate the extraction of distinct features from high-resolution drone orthophotos to support the SVAMITVA Scheme. Leveraging Deep Learning and Computer Vision, we developed a semantic segmentation model capable of identifying 8 distinct classes: Background, Waterbodies, Roads, RCC Roofs, Tiled Roofs, Tin Roofs, Other Roofs, and Utilities. The solution utilizes a U-Net architecture with a ResNet34 backbone, achieving significant accuracy in delineating Abadi (inhabited) land features essential for property record generation.

2. Methodology

2.1 Data Preprocessing & Pipeline

The input dataset consisted of high-resolution Drone TIF images and corresponding shapefiles for villages in Punjab and Chhattisgarh. To prepare this data for the AI model, the following pipeline was established:

- **Tiling Strategy:** Due to the massive size of the original orthophotos (1–3 GB), images were tiled into **512 × 512** patches. Padding was applied to edge tiles to maintain consistent dimensions.
- **Mask Generation (Rasterization):** Ground truth masks were generated from vector shapefiles. Polygon layers were rasterized directly, while linear features (Road Centerlines) and point features (Utilities) were buffered by **3 meters** and **2 meters** respectively to create segmentable polygons.
- **Class Mapping:** Pixels were assigned integer labels (0-7) corresponding to the specific roof types and infrastructure classes defined in the problem statement.
- **Data Split:** The tiled dataset was split into an 80% training set and a 20% validation set to monitor overfitting and ensure generalization.

2.2 Model Architecture

We selected the **U-Net** architecture for this segmentation task. U-Net is an industry-standard for biomedical and geospatial image segmentation due to its ability to localize features precisely using skip connections.

- **Encoder: ResNet34** (pre-trained on ImageNet) to extract deep semantic features.
- **Decoder:** A symmetrical decoder upsamples the features to generate a classification mask.
- **Training Configuration:** The model was trained using the **CrossEntropyLoss** function and the **Adam optimizer** (LR=0.0001) for 10 epochs.

3. Project Outcomes

3.1 Quantitative Evaluation (Accuracy Metrics)

The model was evaluated on a held-out validation set. The performance is detailed below.

Global Metrics:

- **Mean IoU (Intersection over Union):** 0.5732
- **Overall Pixel Accuracy:** 0.9042
- **Macro F1 Score:** 0.6761

Class-wise Performance Table:

Class Label	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Class Name	Background	Waterbody	Road	RCC Roof	Tiled Roof	Tin Roof	Other Roof	Utility
IoU	0.88	0.71	0.62	0.79	0.53	0.79	0.26	0.00
Precision	0.92	0.87	0.87	0.89	0.66	0.88	0.63	0.00
Recall	0.95	0.80	0.69	0.88	0.72	0.88	0.31	0.00
F1-Score	0.94	0.83	0.77	0.89	0.69	0.88	0.42	0.00

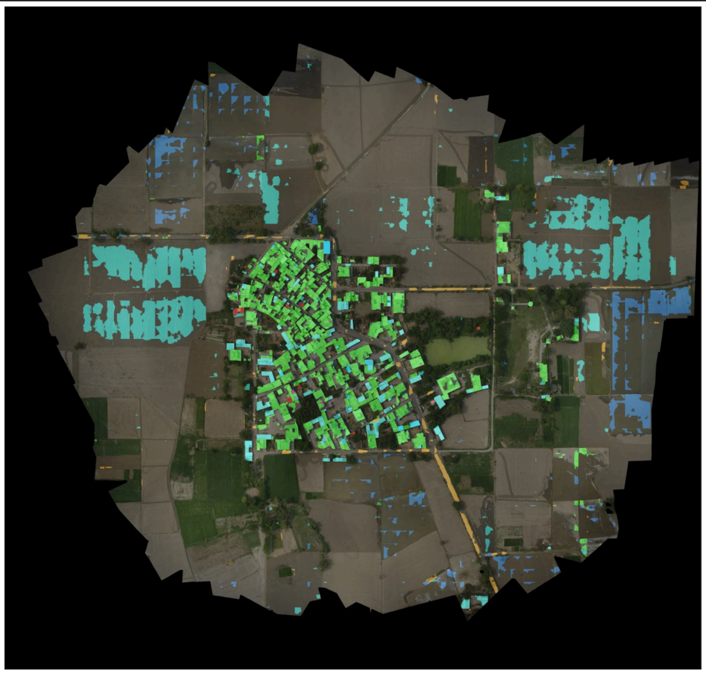
3.2 Qualitative Results (Visual Segmentation)

The following images represent the inference results on the 10 Test Villages (Punjab and Chhattisgarh).

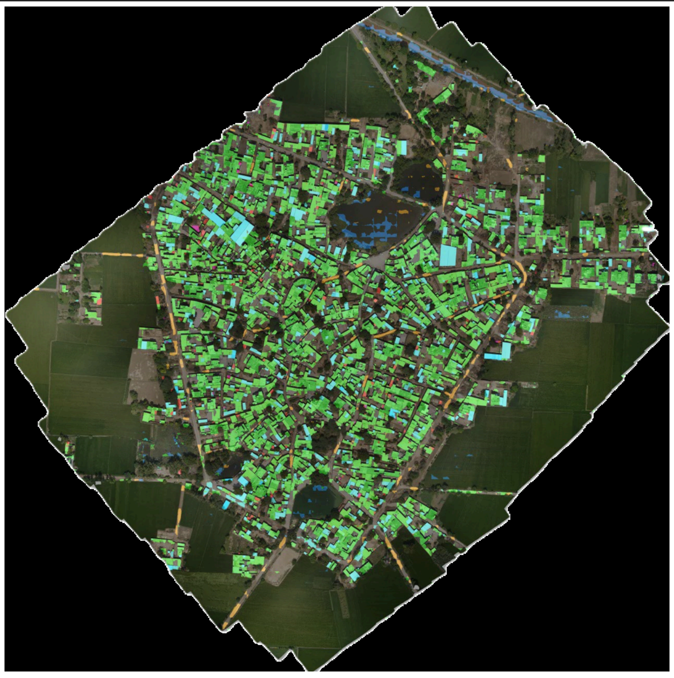
Color Legend:

Waterbody	Road	RCC Roof	Tiled Roof	Tin Roof	Other Roof	Utility	Background

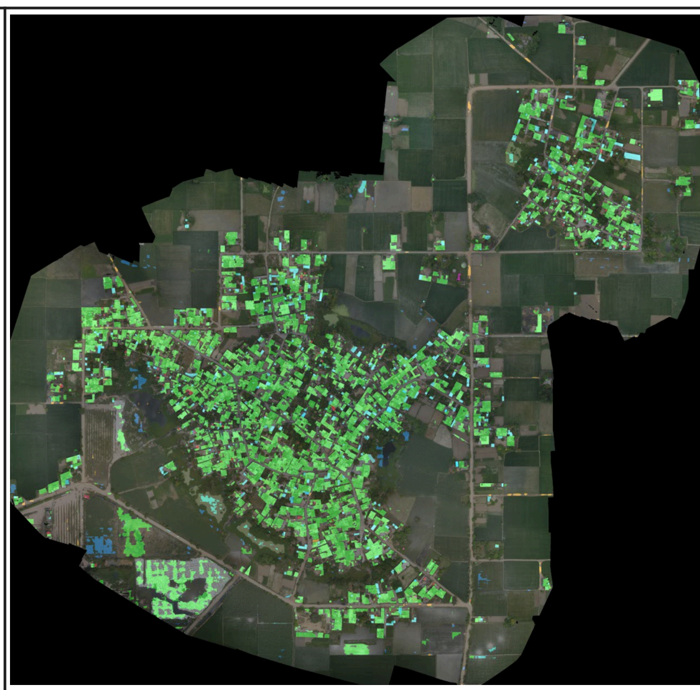
Set 1: Punjab Test Samples



Prediction: ANAITPURA_FATEHGARH SAHIB_32705

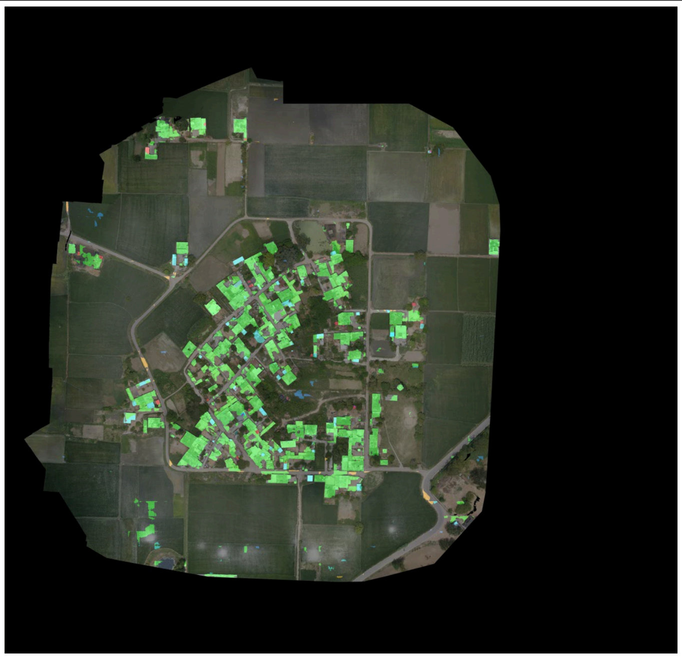


Prediction: BADRA_BARNALA_40044



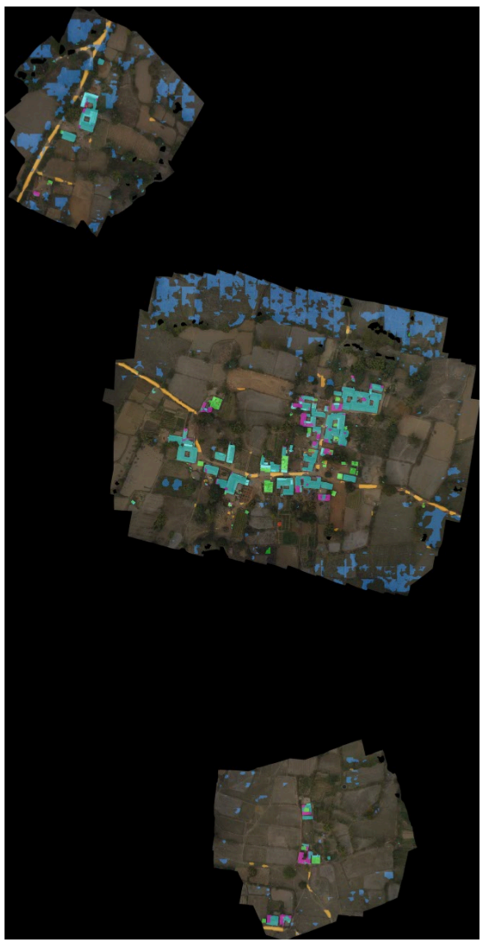
Prediction: BUTTER SIVIYA_AMRITSAR_37780

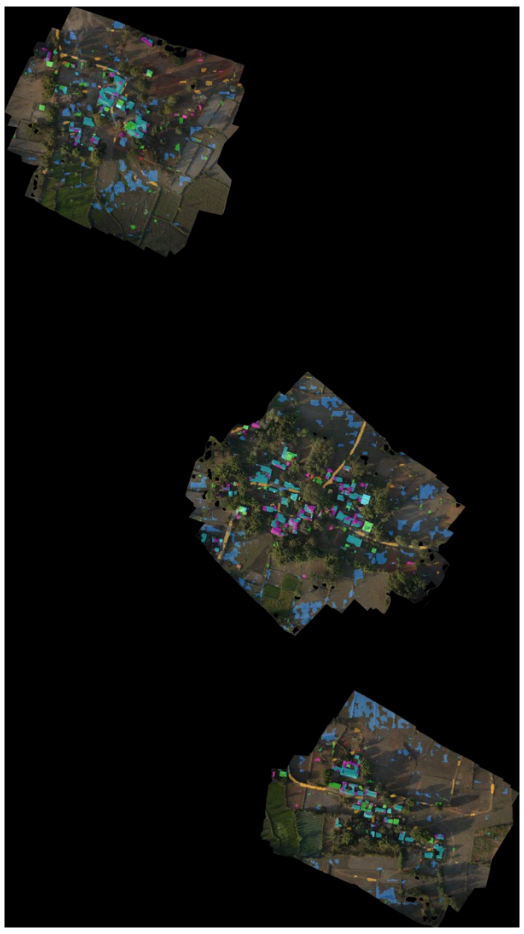
Prediction: DIWANA_BARNALA_40082



Prediction: KARTARPUR_AMRITSAR_37842

Set 2: Chhattisgarh Test Samples





Prediction: CHANABHATA _445476

Prediction: *BAGAI JHARIYA_434310*	Prediction: *BASANTPUR_434297*	
 *Prediction: GUDBHELI_445483*	 *Prediction: PARAGAON_444686*	

4. Observations

Based on the quantitative evaluation and the resulting segmentation maps, the following visual characteristics were observed in the model predictions:

1. **Building Footprint Clarity (RCC & Tin vs. Tiled):**
    - **RCC and Tin Roofs** appear visually sharp and well-defined (Green and Cyan regions). The model successfully delineates the square/rectangular boundaries of these structures with minimal noise, reflecting the high IoU (~0.79) and balanced Precision/Recall.
    - **Tiled Roofs,** conversely, exhibit **"boundary bleeding."** With a low precision of 0.66, the Magenta mask often spills over from the actual roof into surrounding reddish-brown soil or courtyards. The model struggles to distinguish the texture of clay tiles from similar-colored ground, leading to coarser, less accurate shapes.
2. **Road Network Discontinuities:**
    - While the detected road segments are generally correct (High Precision: 0.87), the visual output shows a **fragmented or "dashed" road network** rather than continuous lines.
    - The low Recall (0.69) results in significant gaps in the Orange mask, particularly where tree canopies or shadows obscure the road surface. The model is conservative—it only paints "Road" where it is perfectly visible, failing to infer connections through occluded areas.
3. **The "Invisible" Class (Utilities):** The Yellow mask for **Utility** is virtually non-existent in the visual outputs. Despite the presence of transformers and water tanks in the input images, the extreme class imbalance caused the model to suppress these features entirely, rendering them effectively invisible
4. **Loss of "Other" Roof Structures:** (Red mask) appear patchy and are frequently missed (Recall 0.31).
5. **Background Dominance:** The "Background" (Black) is handled with high fidelity. The model rarely mistakes open fields or empty land for built-up features, resulting in a clean "negative space" around the village settlements.

5. Recommendations for Future Improvements

To bridge the gap between the current prototype and the target accuracy of 95% for the final round, the following strategies are proposed:

1. **Addressing Class Imbalance (Fixing Class 7 & 6):**
    - Loss Function modification: Transition from standard Cross-Entropy Loss to Focal Loss or Weighted Cross-Entropy Loss. This will assign higher penalties to misclassifications of rare classes (Utilities, Other Roofs), forcing the model to learn features for small objects.
    - Oversampling: Implement a sampling strategy that specifically targets tiles containing minority classes during training batches.
2. **Improving Boundary Delineation:**
    - Ensemble Modeling: Combine the current U-Net (ResNet34) with architectures like DeepLabV3+ or PSPNet, which utilize dilation and pyramid pooling to better capture context and improve the segmentation of "Other Roofs" and "Tiled Roofs."
3. **Post-Processing for Roads:**
    - Apply morphological operations (dilation followed by erosion) to reconnect fragmented road segments.
    - Implement a skeletonization algorithm to enforce road continuity in the final vector output.
4. **Data Augmentation:** Introduce aggressive color jittering (brightness/contrast changes) to make the model robust to varying lighting conditions, which may help reduce confusion between Tiled roofs and the background.

6. Conclusion

The Theme-1 prototype successfully demonstrates the application of AI/ML for automated feature extraction under the SVAMITVA scheme. The developed U-Net model achieved a Pixel Accuracy of 90.42% and a Mean IoU of 57.32%, successfully segmenting major topological features such as buildings (RCC/Tin), waterbodies, and roads.

While the model excels at identifying distinct, large-scale structures, the Round 1 analysis highlights a critical need to focus on small-object detection (Utilities) and highly variable classes (Other Roofs). By implementing the recommended advanced loss functions and ensemble techniques, the solution is well-positioned to achieve the high-fidelity mapping required for rural property record generation in the final hackathon stage.